

# Ekman Spirals with Variable Eddy Viscosity

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# 1 Introduction

The Ekman spiral is an important consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the medium, the wind high up in the atmosphere, or wind over the ocean, will not give rise to a flow in the same direction as at the surface. The adjustment of the flow from the surface spirals and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. According to Ekman [1905] the surface wind is at an angle from the wind aloft by  $45^\circ$  [Ekman, 1905]. Due to this turning of the wind the net flux of the wind is perpendicular to the wind aloft which is known as the Ekman transport and is crucial concept in meteorology and oceanography.

However, a surface wind-turning of  $45^\circ$  assumes a constant viscosity with height, which is unrealistic when a turbulent boundary layer is considered. The boundary layer is around 1–2 km thick and is characterised by a presence of turbulence [Wallace and Hobbs, 2006]. The turbulence primarily arises from solar heating, causing instability and turnover, but also from mechanical turbulence from mean wind decreasing closer to the ground due to frictional shear forces. Although turbulence is non-linear, statistical analysis can provide useful first-order approximations, such as the downward flux being negatively and linearly proportional to the local gradient. Using this proportionality constant one can define the eddy viscosity,  $\nu$ , for the boundary layer.

According to the mixing length hypothesis, the eddy viscosity should grow with the distance from the surface as the characteristic scales of the eddies grow. This can furthermore be described by a dimensional analysis of the boundary layer, where a logarithmic relation between the wind speed and height can be obtained [Wallace and Hobbs, 2006]. This showcases that the eddies, and thus the eddy viscosity, grow with height in the boundary layer. However, after a certain altitude, the turbulent boundary layer ceases to exist, and laminar flow takes over, which rapidly decreases the eddy viscosity. Thus, assuming that the eddy viscosity should increase initially from the surface and then vanish at a higher altitude is reasonable. Following this logic, the viscosity should in reality vary with altitude. This would change the surface wind-turning from  $45^\circ$ , and thus also the Ekman transport.

Analytical models have been created to investigate the effect of depth-dependent eddy viscosities for the Ekman spiral. Grisogono [1995] used a Wentzel–Kramers–Brillouin (WKB) approximation, like in quantum mechanics, for the classical Ekman solution, showcasing that smoothly varying viscosity profiles allow for approximate analytical solutions for the entire Ekman height. Bressan and Constantin [2019] used a direct perturbative method on the viscosity in order to obtain a first-order equation for predicting the surface angle of a constant perturbed viscosity. This was later refined by Constantin [2021] to show that the surface angle is a product of the classical angle and a weighted vertical integral of the perturbation. Some exact analytical solutions have also been derived, including a two-layer model by Dritschel et al. [2020], Guan et al. [2023], an exponential decaying profile by Miles [1994] and monotonically increasing polynomial profiles by Ionescu-Kruse [2021], Guan et al. [2023].

However, further insight into the topic can be obtained by directly comparing analytical and numerical solutions. Furthermore, comparing these with these theoretical models with observational data can further help to understand the topic. In this article, a varying viscosity with altitude will be studied under the Ekman equation. This will be done analytically and numerically, and will be compared to reanalysed data. This will give a deeper understanding of the Ekman spiral and provide insight when a surface angle different from  $45^\circ$  can be observed.

## 2 Theory

### 2.1 Problem formulation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming laplace friction, incompressible flow, negligible vertical motion, gives the horizontal

momentum equations as

$$\begin{cases} -fv = -\frac{1}{\rho} \frac{dp}{dx} + \frac{d}{dz} \left( \nu \frac{du}{dz} \right), \\ fu = -\frac{1}{\rho} \frac{dp}{dy} + \frac{d}{dz} \left( \nu \frac{dv}{dz} \right), \end{cases} \quad (1a)$$

$$(1b)$$

where  $u$  and  $v$  are the horizontal velocity components,  $f$  is the Coriolis parameter,  $\rho$  is the fluid density, and  $\nu$  is the viscosity. Far away from the surface, shear flow and friction becomes negligible and the flow will approach geostrophic balance,  $(u, v) \rightarrow (u_g, v_g)$ , where the friction term in Equation 1b can be neglected. In this article, a Coriolis parameter of  $1 \times 10^{-4} \text{ s}^{-1}$  and a zonal geostrophic wind will be used. Subtracting the geostrophic balance from Equation 1b gives

$$\begin{cases} -f(v - v_g) = \frac{d}{dz} \left( \nu \frac{du}{dz} \right), \\ f(u - u_g) = \frac{d}{dz} \left( \nu \frac{dv}{dz} \right). \end{cases} \quad (2a)$$

$$(2b)$$

This problem can be formulated with one equation by introducing a complex velocity as

$$U \equiv u + iv, \quad (3)$$

and a corresponding complex geostrophic velocity. Using Equation 2b with the complex velocities  $U$  and  $U_g$  the coupled equations can be combined into the Ekman equation

$$\frac{d}{dz} \left( \nu \frac{dU}{dz} \right) = if(U - U_g), \quad (4)$$

with the boundary conditions

$$U(0) = 0, \quad (5a)$$

$$U(z \rightarrow \infty) = U_g. \quad (5b)$$

The surface wind angle can be determined by taking the argument of the infinitesimal change of the wind at the ground,

$$\theta = \arg \left( \frac{dU}{dz} \Big|_{z=0} \right). \quad (6)$$

The Ekman transport is defined as the mass flux per unit width for the entire Ekman spiral, which can be defined as

$$T = \int_0^\infty [U - U_g] dz, \quad (7)$$

since the transport represents the net movement of the entire fluid column offset from the geostrophic wind aloft. The angle of the Ekman transport can consequently be determined by taking the argument,  $\theta_T = \arg(T)$ .

## 2.2 The vertical momentum flux

The Ekman equation can be reformulated by introducing a complex vertical momentum flux as

$$\tau = \nu \frac{dU}{dz}. \quad (8)$$

Using this definition, and differentiating Equation 4 gives

$$\boxed{\frac{d^2 \tau}{dz^2} = i\kappa^2 \tau}, \quad (9)$$

where

$$\kappa^2 \equiv \frac{f}{\nu}. \quad (10)$$

Here, an inverse decay scale  $\kappa$  has been introduced to the system. Equation 9 is similar to equations such as the Helmholtz equation and the time-independent Schrödinger equation, where  $\kappa$  becomes analogous to the potential. The new equation also has a clear physical interpretation, as it describes the vertical structure of the momentum flux in the Ekman problem. Using Equation 4 and 8, the boundary conditions become

$$\left. \frac{d\tau}{dz} \right|_{z=0} = -ifU_g, \quad (11a)$$

$$\tau(z \rightarrow \infty) = 0. \quad (11b)$$

Since the viscosity is always a positive real value the surface angle can also be expressed in terms of  $\tau$  as

$$\theta = \arg[\tau(z=0)]. \quad (12)$$

Solving for the Ekman transport becomes easier by using the momentum flux, as it directly reduces the integral of the Ekman transport. Inserting Equation 4 into 8 it follows that  $\frac{d\tau}{dz} = if(U - U_g)$ . Using the boundary condition that the momentum flux vanishes at infinity the Ekman transport reduces to

$$T = \int_0^\infty [U - U_g] dz = \frac{i}{f} \tau(z=0), \quad (13)$$

which is proportional to the vertical momentum flux at the surface. Thus, the surface stress at the ground is proportional to the Ekman transport of the fluid column.

## 2.3 The classical solution

The classical solution to the Ekman equation is obtained when the viscosity is assumed to be constant with altitude,  $\nu = \text{constant}$ , and was found by Ekman [1905]. In this case the classical Ekman spiral solution is

$$\tau(z) = \nu U_g \frac{1+i}{h_{\text{Ek}}} e^{-\frac{1+i}{h_{\text{Ek}}} z}, \quad (14)$$

where

$$h_{\text{Ek}} \equiv \sqrt{\frac{2\nu}{f}} \quad (15)$$

is the Ekman height scale. The surface angle of the classical Ekman theory gives  $45^\circ$ , since the solution is equally real and imaginary at  $z = 0$ . The transport yields an angle of  $90^\circ$  from the surface wind-turing, meaning that the transport and surface wind is perpendicular.

## 2.4 The two layer model

The next natural step from a constant viscosity profile is to assume two layers of constant viscosity in each layer. The lower level is labelled as having the viscosity  $\nu_1$  up until a height of  $H$ , where the next viscosity of  $\nu_2$  takes over. From this assumption a superimposed solution similar to Equation 14 is obtained as

$$\begin{cases} \tau_1(z) = A_1 e^{\frac{1+i}{h_{\text{Ek}1}} z} + B_1 e^{-\frac{1+i}{h_{\text{Ek}1}} z}, \\ \tau_2(z) = B_2 e^{-\frac{1+i}{h_{\text{Ek}2}} z} \end{cases} \quad (16a)$$

$$(16b)$$

where the Ekman height scales are defined for each layer like in Equation 15. To solve for the prefactors above one has to use that the velocity and momentum flux is continuous, which gives the jump conditions

$$U_1 = U_2 \quad \text{at } z = H, \quad (17a)$$

$$\tau_1 = \tau_2 \quad \text{at } z = H. \quad (17b)$$

Using the jump conditions together with the boundary conditions in Equation 5 an expressions for the prefactors is obtained as

$$A_1 = -\nu_1 U_g \frac{1+i}{h_{\text{Ek1}}} \cdot \frac{1-\gamma}{1-\gamma + e^{2(1+i)\varphi} (1+\gamma)}, \quad (18a)$$

$$B_1 = \nu_1 U_g \frac{1+i}{h_{\text{Ek1}}} \cdot \frac{(1+\gamma) e^{2(1+i)\varphi}}{1-\gamma + e^{2(1+i)\varphi} (1+\gamma)}, \quad (18b)$$

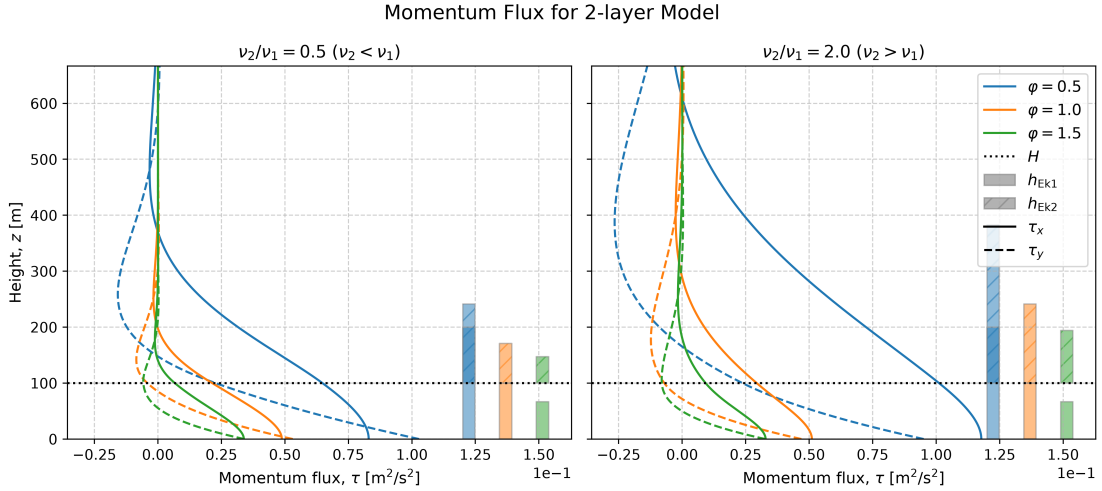
$$B_2 = \nu_2 U_g \frac{1+i}{h_{\text{Ek2}}} \cdot \frac{2e^{(1+i)\varphi}(1+\gamma^{-1})}{1-\gamma + e^{2(1+i)\varphi} (1+\gamma)}, \quad (18c)$$

$$\text{where } \gamma = \sqrt{\frac{\nu_2}{\nu_1}} \quad (18d)$$

and where  $\varphi$  is the non-dimensional layer thickness defined as

$$\varphi = \frac{H}{h_{\text{Ek1}}}. \quad (19)$$

This provides a complete expression for the vertical momentum flux in the two-layer model, as shown in Figure 1. The figure illustrates how the vertical momentum flux varies with altitude for different viscosity ratios and non-dimensional heights,  $\varphi$ . For an increasing viscosity with height, a larger surface stress in the zonal direction compared to the meridional direction is observed. Conversely, for a smaller upper-layer viscosity, a larger meridional surface stress compared to the zonal surface stress is evident. Observing the interface,  $H$ , reveals that for larger  $\varphi$ , the model behaves more like a single-layer model, as the second layer begins at a height where the Ekman spiral has already decayed. Interestingly, smaller values of  $\varphi$  yield a larger difference between the zonal and meridional wind stress.



**Figure 1:** The vertical structure of the momentum flux and the inverse decay scales for the two layer model. The momentum flux is dependent on the dimensionless parameter  $\varphi = H/h_{\text{Ek1}}$  and the ratio of the viscosities. The corresponding Ekman heights and interface are shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow  $U_g = 10 \text{ m s}^{-1}$ , a Coriolis parameter of  $f = 1 \times 10^{-4} \text{ s}^{-1}$  and an intersection height of  $H = 100 \text{ m}$ . The lower viscosities, derived from  $H$  and  $\varphi$ , are  $\nu_1 = 2 \text{ m}^2/\text{s}$ ,  $\nu_1 = 5 \times 10^{-1} \text{ m}^2/\text{s}$ , and  $\nu_1 = 2.22 \times 10^{-1} \text{ m}^2/\text{s}$  in both figures. In the left figure,  $\nu_2$  used half the value of the lower viscosities, and in the right figure, it used double of that value.

By defining the constant

$$R = \frac{1-\gamma}{1+\gamma}. \quad (20)$$

and using the derived solution for the vertical momentum flux, the surface wind angle becomes

$$\theta = \arg [(A_1 + B_1)] \quad (21)$$

$$= 45^\circ + \arg \left[ \frac{e^{(1+i)\varphi} - R e^{-(1+i)\varphi}}{e^{(1+i)\varphi} + R e^{-(1+i)\varphi}} \right], \quad (22)$$

which can be observed as a function of  $\varphi$  in Figure 2. The solutions for different viscosity ratios are shown, half of which have a higher upper-level viscosity and the other half have a lower upper-level viscosity. One can observe a convergence towards the classical Ekman value of  $45^\circ$  when  $\varphi \rightarrow \infty$  for all the viscosity ratios. This can be attributed to the high altitude interface  $H$  being above the spiral, which reduces the problem to a one layer model.

The zero limit is, however, dependent on the viscosity ratio. If the ratio is finite and fixed, the surface angle approaches  $45^\circ$  as  $H \rightarrow 0$ , again reducing the problem to a one-layer model. But if  $\varphi$  is very small while the viscosity ratio varies, the angle approaches  $0^\circ$  if  $\nu_2/\nu_1 \rightarrow \infty$ , or an angle of  $90^\circ$  if  $\nu_2/\nu_1 \rightarrow 0$ . **The surface is thus sensitive to the viscosity ratio when  $\varphi$  is small, as the surface angle can be either above or below  $45^\circ$  depending on the viscosity ratio. When  $\varphi \ll 1$ , the solution behaves approximately as a linear function between 0 and  $H$ , allowing the jump conditions to determine the surface angle. Observing Equation 16, one can observe:**

$$\left. \frac{d\tau}{dz} \right|_{z=H} \approx -\frac{1+i}{h_{\text{Ek2}}} \tau(H), \quad (23)$$

which puts a constraint on  $\tau(H)$  as  $h_{\text{Ek2}}$  becomes very small or very large. If the viscosity ratio is very small,  $\gamma \ll \varphi$ , the the momentum flux at the interface is very small, due to the small upper Ekman height  $h_{\text{Ek2}} \ll H$  balancing a finite change in momentum flux. Taylor expanding the momentum flux at the interface gives

$$\tau(H) = \tau(0) + H\tau'(0) + \mathcal{O}(H^2). \quad (24)$$

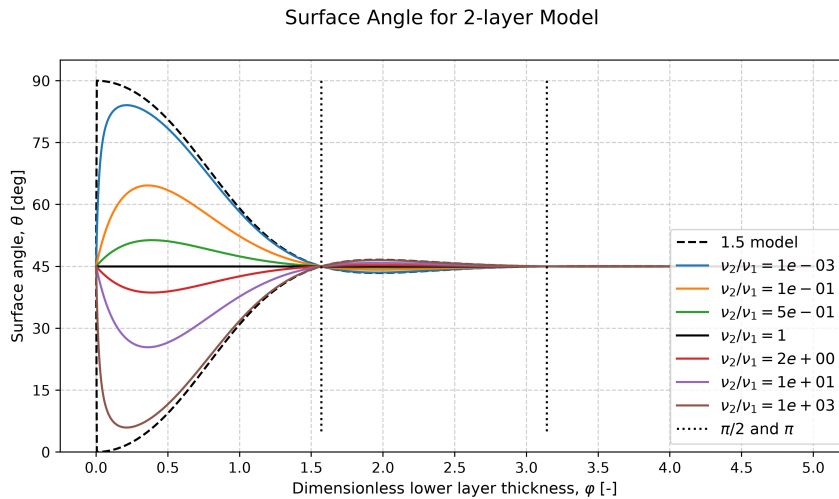
Using Equation 11a and neglecting the small terms gives

$$\tau(0) \approx iU_g H \implies \arg[\tau(0)] \approx 90^\circ. \quad (25)$$

Similarly when  $\varphi \ll 1$  and  $\gamma \gg \varphi$ , the change in the momentum flux has to become small, due to large upper Ekman height,  $h_{\text{Ek2}} \gg H$  so that the change becomes small to balance the finite momentum flux. Integrating Equation 9 gives

$$\tau'(H) - \tau'(0) = i\frac{fH}{\nu_1}\tau(0) + \mathcal{O}(H^2) \implies \tau(0) \approx \frac{\nu_1 U_g}{fH} \implies \arg[\tau(0)] \approx 0^\circ. \quad (26)$$

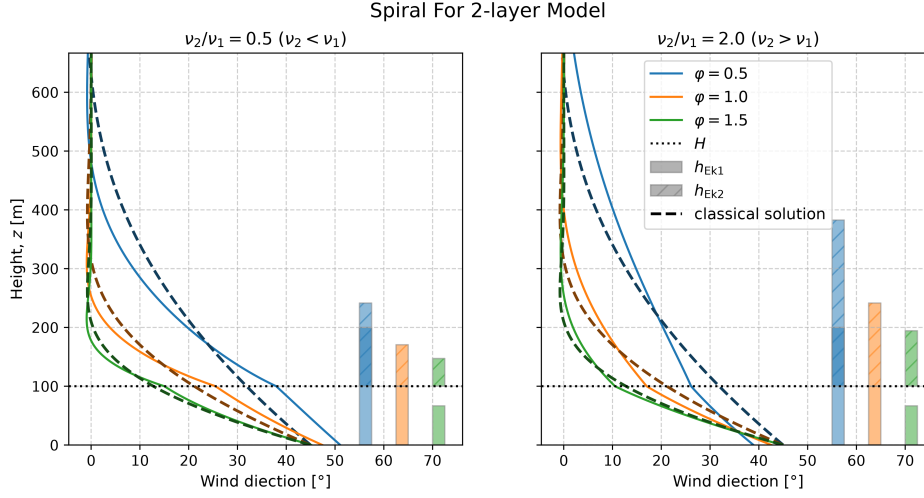
In the case of weaker upper viscosity the perpendicular angle can be interpreted as the entire rotation occurring at the surface since the viscosity becomes negligible aloft. In the opposite case, a weaker lower viscosity allowing minimal spiralling, since the absence stress cannot force a rotation.



**Figure 2:** The surface angle as a function of the dimensionless layer thickness  $\varphi = H/h_{\text{Ek1}}$  for the two-layer model. Different coloured curves correspond to different layer viscosity ratios. The classical Ekman angle of  $45^\circ$  is shown as a reference.

The full vertical spiral of the two-layer model can be obtained by calculating the wind direction at each height. This spiralling compared to the classical Ekman theory can be observed in Figure 3. If  $\varphi$  is large,

the most of the spiral is in the lower layer, and therefore close to the classical Ekman theory. In the other cases it is seen that the angle overshoots in the layer with high viscosity and undershoots in layer with low viscosity compared to the classical solution.



**Figure 3:** The vertical spiral of the velocities for the two layer model together with the corresponding Ekman heights. The wind direction is dependent on the dimensionless parameter  $\varphi = H/h_{\text{Ek1}}$ . The classical Ekman solution (when  $\nu_{\text{const}} = \nu_1$ ) is shown as a reference. The parameters used the same values as in Figure 1.

## 2.5 The exponential decaying viscosity model

Another viscosity profile that can provide a fully analytical solution is an exponentially decaying viscosity with altitude. Here the viscosity is set to decay from an initial viscosity of  $\nu_0$ , with a decay rate of  $k$  as  $\nu(z) = \nu_0 \exp(-kz)$ . Solving Equation 9 with this viscosity profile can be done by introducing the new variable

$$t = 2(1+i) \varphi_{\text{exp}} e^{\frac{kz}{2}}. \quad (27)$$

The non-dimensional parameter  $\varphi_{\text{exp}}$  resembles the non-dimensional layer thickness from the two-layer model, but defined as

$$\varphi_{\text{exp}} = \frac{1}{k h_{\text{Ek0}}} = \frac{1}{k} \sqrt{\frac{f}{2\nu_0}}. \quad (28)$$

Using these newly defined variables Equation 9 transforms into

$$t^2 \frac{d^2 \tau}{dt^2} + t \frac{d\tau}{dt} - t^2 \tau = 0, \quad (29)$$

which is a differential equation with a standard solution. The standard solution is a modified Bessel equation as

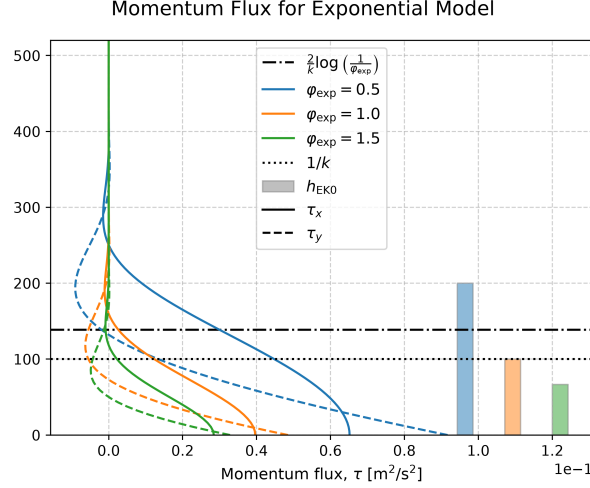
$$\tau(t) = C_1 I_0(t) + C_2 K_0(t), \quad (30)$$

where  $C_1$  and  $C_2$  are constants, while  $I_0(t)$  and  $K_0(t)$  are modified Bessel functions of the first and second kind of order zero, respectively. By applying the previously defined boundary conditions from Equation 11 the two constants can be found. Applying the geostrophic boundary condition from Equation 11b shows that  $C_1 = 0$  since  $I_0(t)$  does not go to zero when  $t \rightarrow \infty$  (which occurs when  $z \rightarrow \infty$ ). Applying the surface boundary condition from Equation 11a an expression for the second constant can be found. Transforming the solution to depend on the variable  $z$  gives

$$\tau(z) = f U_g \frac{1+i}{h_{\text{Ek0}}} \frac{K_0(2[1+i]\varphi_{\text{exp}} e^{kz/2})}{K_1(2[1+i]\varphi_{\text{exp}})}. \quad (31)$$

This solution for the vertical momentum flux in the exponential model is shown in Figure 4 for different non-dimensional heights,  $\varphi_{\text{exp}}$ . For smaller  $\varphi_{\text{exp}}$ , the solution is more localized toward the surface, whereas larger  $\varphi_{\text{exp}}$  produces a more extended spiral. This implies that for larger  $\varphi_{\text{exp}}$ , the viscosity changes very





**Figure 4:** The vertical structure of the momentum flux for exponentially decreasing viscosity together with the corresponding Ekman height and inverse decay rate. The momentum flux is dependent on the dimensionless parameter  $\varphi_{\text{exp}} = 1/k h_{\text{Ek}0}$ . The solution assumed a purely zonal geostrophic flow  $U_g = 10 \text{ m s}^{-1}$ , a Coriolis parameter of  $f = 1 \times 10^{-4} \text{ s}^{-1}$ , a decay rate of  $k = 1 \times 10^{-2} \text{ m}^{-1}$  and initial viscosities of  $\nu_0 = 12.5 \text{ m}^2 \text{ s}^{-1}$ ,  $\nu_0 = 0.5 \text{ m}^2 \text{ s}^{-1}$  and  $\nu_0 = 0.03125 \text{ m}^2 \text{ s}^{-1}$ .

little, causing the spiral to behave as in the one-layer model. In contrast, smaller  $\varphi_{\text{exp}}$  indicates a rapid viscosity decrease from the ground.

Using Equation 31 the surface angle of this model is

$$\theta = 45^\circ + \arg \left[ \frac{K_0 (4 [1 + i] \varphi_{\text{exp}})}{K_1 (4 [1 + i] \varphi_{\text{exp}})} \right], \quad (32)$$

where  $K_1$  is the modified Bessel function of the second kind of order one. Equation 32 is plotted in Figure 5. As this model only provides scenarios where the viscosity decreases from the ground, all angles are greater than the reference angle of  $45^\circ$  (as predicted by the two layer model). One can observe a similar limit as in Figure 2 when  $\varphi_{\text{exp}} \rightarrow 0$  for  $\nu_2 = 0$ , as Equation 32 also approaches  $90^\circ$ . The same limit is also true for  $\varphi_{\text{exp}} \rightarrow \infty$ , which gives  $\theta \rightarrow 45^\circ$ , although this exponential solution converges slightly slower than the two-layer solution. The reasoning behind this convergence is that the lower decay rates makes the solution behave more like the classical Ekman spiral with constant viscosity.

The opposite case when  $\varphi_{\text{exp}} \ll 1$  allows for Taylor expansion of  $\tau(z)$  about  $z = 0$  as

$$\tau(z) \approx \tau_0 + \tau'(0)z + \tau''(0)\frac{z^2}{2} \quad (33a)$$

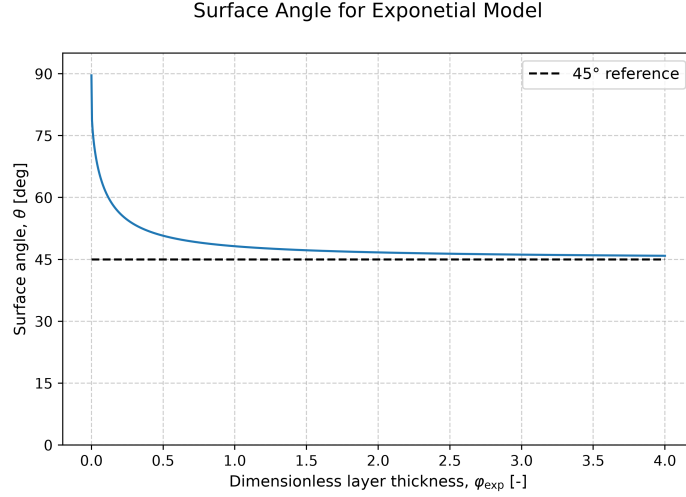
$$\approx \tau_0 - \underbrace{iU_g z}_{\text{linear}} + \underbrace{\frac{if\tau_0}{2\nu_0} z^2}_{\text{curvature}}, \quad (33b)$$

$$(33c)$$

at the characteristic height  $z = 1/k$  gives

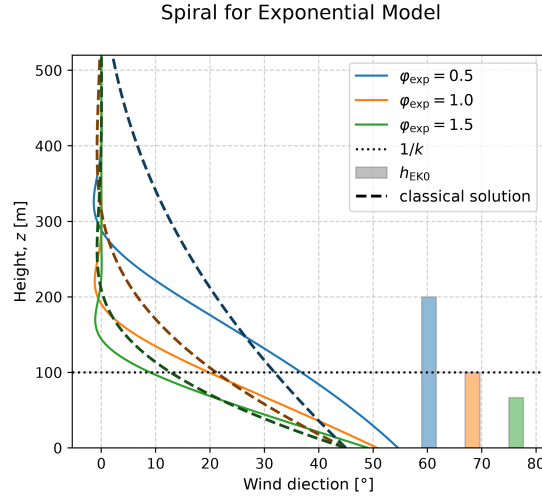
$$\tau\left(\frac{1}{k}\right) \approx \tau_0 - \frac{iU_g}{k} + \varphi_{\text{exp}}^2 \tau_0. \quad (34)$$

Thus, as  $\varphi_{\text{exp}}^2 \rightarrow 0$  the curvature term becomes negligible. Furthermore, for an exponentially viscosity increases,  $f/\nu(z)$  decreases exponentially with height. Hence, the curvature term becomes even smaller higher up, allowing for leading linear behaviour. Matching its behaviour at the surface, characteristic height, and  $\tau \rightarrow 0$  as  $z \rightarrow \infty$  requires the imaginary linear term to mainly balance  $\tau_0$ . Thus the surface momentum flux has to be imaginary to the leading order, providing a perpendicular surface angle of  $90^\circ$  when  $\varphi_{\text{exp}} \rightarrow 0$ .



**Figure 5:** Surface angle for the exponentially decaying viscosity model as a function of the dimensionless parameter  $\varphi_{\text{exp}} = H/h_{\text{Ek}0}$ . The same parameters as in Figure 4 was used.

The wind directions of the exponential model is shown in Figure 4, compared to the classical Ekman theory. The angles are higher than the classical solution at the surface but decreases more rapidly with height, since the viscosity decreases, following the logic presented in the two-layer model.



**Figure 6:** The vertical spiral for the exponentially decreasing viscosity model together with the Ekman height and the inverse decay rate. The wind direction is dependent on the dimensionless parameter  $\varphi_{\text{exp}} = 1/k h_{\text{Ek}0}$ . The classical Ekman solution (for  $\nu_0$ ) is shown as a reference. The parameters had the same values as in Figure 4.

## 2.6 The simplified upper boundary condition model

There is difficulty solving Equation 9 due to the altitude dependence of the viscosity profile  $\nu(z)$ . Analytical solutions therefore exist only for specific viscosity profiles that allow the equation to be transformed into a solvable form. Equation 9 is furthermore also difficult to solve numerically, as the upper boundary condition is asymptotic. This will put limits on the viscosity profiles to study, as the viscosity cannot increase with altitude continuously. In the physical world, this makes sense, since we assume that high up in the atmosphere (or deep down in the ocean), the flow becomes laminar and the turbulent viscosity small. I.e. that the turbulent boundary layer has a boundary. However, by limiting ourselves to only decreasing viscosity profiles, the real-world differential equation becomes impossible to solve.

A solution to solve this problem is to introduce a simplified upper boundary condition. For the two-layer

model, this is reduced to a 1.5 layer model, where we assume that the upper-level viscosity is negligible ( $\nu_2 \approx 0$ ), which corresponds to the upper dashed curve in Figure 2. This also makes physical sense, since we know that both the atmosphere and the ocean consist of a boundary layer where turbulent mixing takes place. Allowing the viscosity to vary freely in the lower layer reduces the problem to a simplified upper boundary condition model. This will allow viscosity profiles that do not converge at  $z \rightarrow \infty$ , since the new domain is between  $z \in [0, H]$ . This modifies the boundary conditions in Equation 11 as

$$\left. \frac{d\tau}{dz} \right|_{z=0} = -ifU_g, \quad (35a)$$

$$\tau(z = H) = 0. \quad (35b)$$

Here,  $H$  is the height of the boundary layer, and the upper boundary condition was argued from the jump condition in the two-layer model, as seen in Equation 17b.

### 2.6.1 Simple viscosity profiles for the SUBC model

Using the the simplified upper boundary condition, SUBC model, simple viscosity profiles can be studied and evaluated numerically. The easiest, except for the constant viscosity, is a linearly varying viscosity with height as

$$\nu_{\uparrow}(z) = \frac{\nu_{\max}}{H} z, \quad (36a)$$

$$\nu_{\downarrow}(z) = \frac{\nu_{\max}}{H} (H - z), \quad (36b)$$

where  $\nu_{\max}$  is the maximal viscosity. For both of these profiles a dimensionless vertical coordinate can be defined as

$$\tilde{z} = \frac{z}{H} \implies \begin{cases} \nu_{\uparrow}(z) = \nu_{\max} \tilde{z} \\ \nu_{\downarrow}(z) = \nu_{\max} (1 - \tilde{z}). \end{cases} \quad (37)$$

Using these linear viscosity profiles, together with the new dimensionless vertical coordinate, Equation 9 is rewritten for the two profiles as

$$\frac{d^2\tau}{d\tilde{z}^2} = i \underbrace{\frac{1}{\tilde{z}}}_{\equiv 1/\tilde{\nu}_{\uparrow}} \underbrace{\frac{H^2 f}{\nu_{\max}}}_{=2\varphi^2} \tau, \quad (38a)$$

$$\frac{d^2\tau}{d\tilde{z}^2} = i \underbrace{\frac{1}{1-\tilde{z}}}_{\equiv 1/\tilde{\nu}_{\downarrow}} \underbrace{\frac{H^2 f}{\nu_{\max}}}_{=2\varphi^2} \tau. \quad (38b)$$

Above the dimensionless thickness  $\varphi$  was again introduced following the logic from before.

Another simple profile to evaluate is a parabolic increasing viscosity as

$$\nu_p = \frac{4\nu_{\max}}{H^2} (H - z) z. \quad (39)$$

Again, using the non-dimensional height coordinate  $\tilde{z}$  and the dimensionless thickness  $\varphi$ , Equation 9 is rewritten as

$$\frac{d^2\tau}{d\tilde{z}^2} = i \underbrace{\frac{1}{4\tilde{z}[1-\tilde{z}]}}_{\equiv 1/\tilde{\nu}_p} \underbrace{\frac{H^2 f}{\nu_{\max}}}_{=2\varphi^2} (U - U_g). \quad (40)$$

Since the height coordinate is normalized for the height  $H$ , the dimensionless thickness simply becomes the same as the inverse of the normalized Ekman scale height, defined as

$$\varphi = \frac{H}{h_{\text{Ek}}} = \frac{1}{\tilde{h}_{\text{Ek}}}. \quad (41)$$

### 3 Methodology

#### 3.1 Numerical method

Equation 9 was solved numerically as a boundary value problem using the `solve_bvp` routine available in SciPy [Virtanen et al., 2020]. For the SUBC model, the equation was shown to be expressed in terms of a dimensionless layer thickness by using an height-normalized eddy viscosity profile,  $\tilde{\nu}(z)$ . Separating Equation 38a, 38b and 40 into zonal and meridional components yields

$$\frac{d^2\tau_x}{dz^2} = -\frac{1}{\tilde{\nu}(z)}2\varphi^2\tau_y, \quad (42a)$$

$$\frac{d^2\tau_y}{dz^2} = \frac{1}{\tilde{\nu}(z)}2\varphi^2\tau_x. \quad (42b)$$

To apply the numerical boundary value solver, the equations were rewritten as a first-order system by introducing the variables

$$\tau'_x = \frac{d\tau_x}{dz}, \quad \tau'_y = \frac{d\tau_y}{dz}. \quad (43)$$

Expanding Equation 42b into four coupled first-order equations gives

$$\begin{cases} \frac{d\tau_x}{dz} &= \tau'_x \\ \frac{d\tau'_x}{dz} &= -\frac{2\varphi^2\tau_y}{\tilde{\nu}(z)} \\ \frac{d\tau_y}{dz} &= \tau'_y \\ \frac{d\tau'_y}{dz} &= \frac{2\varphi^2\tau_x}{\tilde{\nu}(z)}, \end{cases} \quad (44)$$

with the boundary conditions

$$\tau'_x(\varphi = 0) = 0, \quad \tau_x(\varphi = 1) = 0, \quad (45a)$$

$$\tau'_y(\varphi = 0) = -fu_g, \quad \tau_y(\varphi = 1) = 0. \quad (45b)$$

If the viscosity profile,  $\tilde{\nu}$ , includes a zero value in the domain, then the coupled Equation 44 becomes singular. We therefore add a constant background viscosity,  $\epsilon$  as

$$\hat{\nu} \rightarrow \hat{\nu} + \epsilon, \quad \text{where } \epsilon = \frac{\nu_{\text{background}}}{\nu_{\text{max}}} \quad (46)$$

and  $\hat{\nu}/\epsilon \gg 1$  is assumed. This background viscosity was thus applied for the linear decreasing, linear increasing, and the parabolic model since they have zero viscosity at the surface (or at the top).

The SUBC model equations were solved iteratively on a vertical grid. The solver required an initial guess, which was chosen based on the initial setup of the system. The initial guess was therefore set as the classical solution for a constant viscosity. The final solution was found when the iterative solution reached a steady state for the differential equations with the imposed boundary conditions.

#### 3.2 Analysis of wind-turning angles from ERA5

Reanalysed atmospheric data for different pressure levels were downloaded from the ERA5 dataset [Copernicus Climate Change Service, 2019] in order to observe the real-world surface wind angles. The data used an annual mean wind dataset for the year 2025. The analysis was restricted to the northern mid-latitudes in order to capture the Ekman spiral from the presence of the jet stream. In order to only analyse land points, a land mask was used from the package `Global_Land_Mask`, which is based on data from NOAA National Centers for Environmental Information [1999]. To avoid mountainous or very hilly terrain, an elevation dataset from NOAA National Centers for Environmental Information [2022] was used in order to dismiss points with a very steep linear slope. For the remaining grid points, the horizontal wind vectors were analysed. The surface wind vectors were compared to the wind at the location of the highest wind speed, i.e., the jet stream. The angle between these two vectors was computed as the signed angle between them by using the cross product. To avoid poorly defined angles, surface wind speeds below a threshold were discarded. The criteria for the data points can be summarised as follows:

- Northern mid-latitude band between  $[30^\circ, 60^\circ]$ .
- Land points only.
- Flat points with a linear slope below  $1000 \text{ m km}^{-1}$ .
- Surface wind speeds above  $1 \text{ m s}^{-1}$ .

Lastly, a circular mean was calculated for the different times in order to obtain a average value for the temporal series. Thus, a geographical map together with angle distributions for the surface wind angles were obtained.

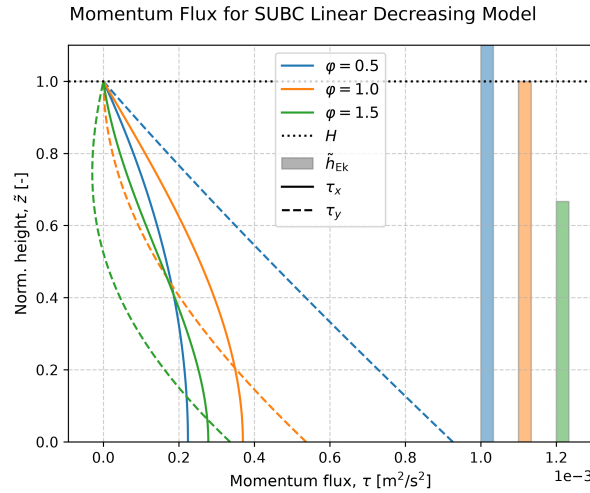
## 4 Result

### 4.1 The theoretical surface angles

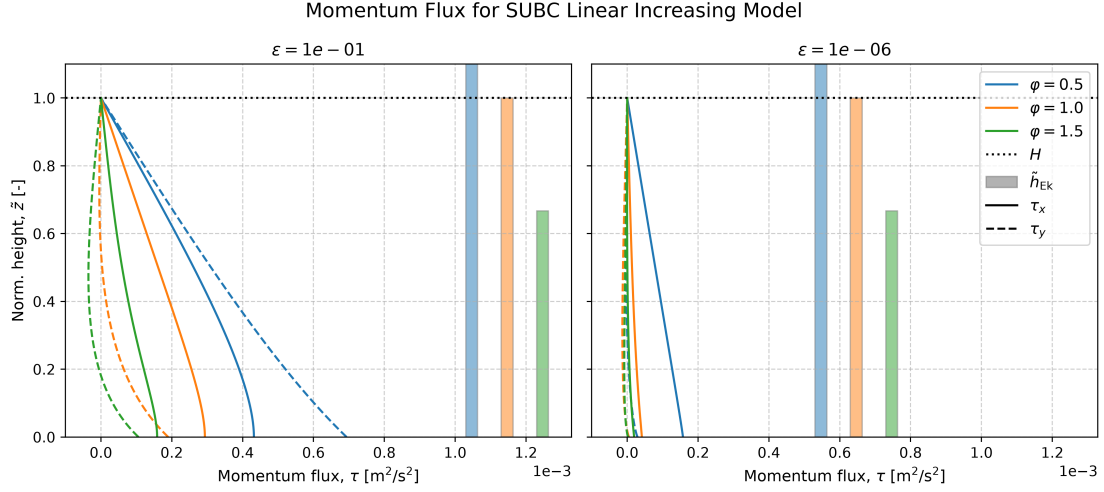
#### 4.1.1 The vertical structure of the momentum flux

The vertical profiles from the numerical SUBC model can be viewed in Figure 7–9 together with the normalized Ekman height for the different layers. An important note is that when  $\varphi < 1$  then the Ekman heights become larger than the limit  $H$ , and the spiral therefore extends over a large height. In contrast, if  $\varphi \gg 1$ , then the Ekman height is much smaller than the layer depth,  $h_{\text{EK}} \ll H$ , such that the spiral is closer to the surface as  $z \ll H$ . Another important observation in Figure 7–9 is that for  $\varphi < 1$ , almost linear profiles are observed. This supports the claim above that when  $\varphi \ll 1$ , then  $\tau(z)$  in Equation 9 behaves as a linear function.

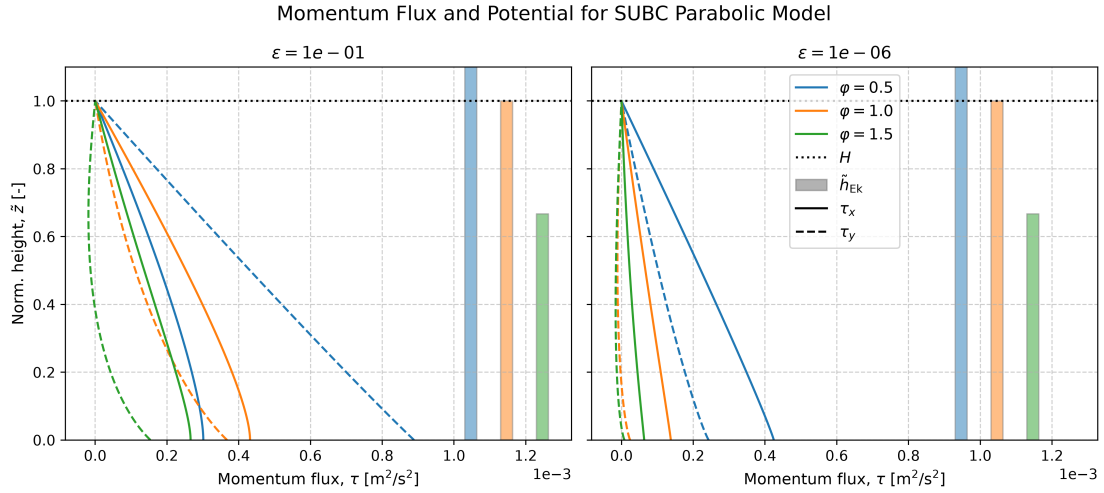
Since the surface wind angle is calculated by comparing the initial change of the viscosity, both the linear increasing and the parabolic models are dependent on the background viscosity  $\epsilon$ . The effect of varying  $\epsilon$  can be observed in Figure 8–9. However for the linear decreasing model, no major effect could be observed by varying the background viscosity. Thus, a background viscosity of  $1 \times 10^{-6}$  was used for this specific model.



**Figure 7:** Numerically computed vertical structure for linearly decreasing viscosity. The momentum flux is dependent on the dimensionless parameter  $\varphi = \tilde{h}_{\text{EK}}^{-1}$ . The solution assumed a purely zonal geostrophic flow  $U_g = 10 \text{ m s}^{-1}$  and a Coriolis parameter of  $f = 1 \times 10^{-4} \text{ s}^{-1}$ .



**Figure 8:** Numerically computed vertical structure for linearly increasing viscosity. The momentum flux is dependent on the dimensionless parameter  $\varphi = \tilde{h}_{\text{Ek}}^{-1}$  and the background viscosity  $\epsilon$ . The solution assumed a purely zonal geostrophic flow  $U_g = 10 \text{ m s}^{-1}$  and a Coriolis parameter of  $f = 1 \times 10^{-4} \text{ s}^{-1}$ .

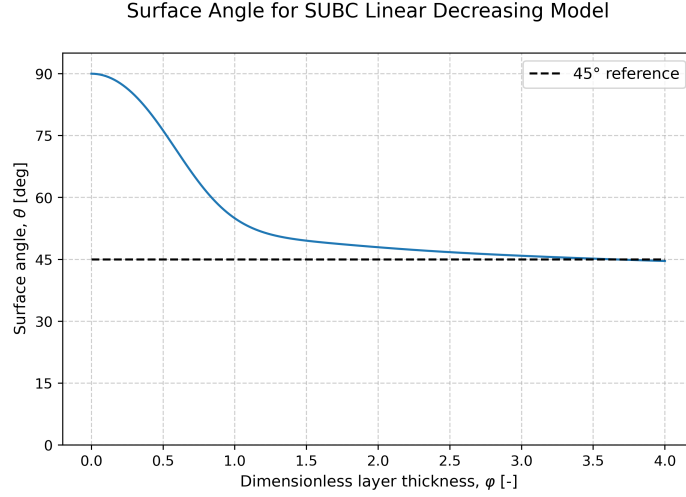


**Figure 9:** Numerically computed vertical structure for parabolic viscosity. The momentum flux is dependent on the dimensionless parameter  $\varphi = \tilde{h}_{\text{Ek}}^{-1}$  and the background viscosity  $\epsilon$ . The solution assumed a purely zonal geostrophic flow  $U_g = 10 \text{ m s}^{-1}$  and a Coriolis parameter of  $f = 1 \times 10^{-4} \text{ s}^{-1}$ .

#### 4.1.2 The surface angles

Figure 10–12 show the surface angles from the numerical SUBC models. Similarities between these models, the decreasing two-layer model, and the exponentially decaying model can be observed. They all have a lower limit of  $90^\circ$  and an upper limit of  $45^\circ$ . The reason for the similarity between the exponential model and the increasing two layer model, is that the SUBC model is based on the decreasing two layer model, as  $\nu = 0$  above the height  $H$ . Again the linear behaviour of  $\tau(z)$  when  $\varphi \ll 1$  yields a surface angle of  $90^\circ$ , while the more constant viscosity when  $\varphi \gg 1$  yields a surface angle of  $45^\circ$ . Thus, in the limit when  $\varphi \rightarrow 0$  the Ekman height becomes infinitely larger than  $H$  providing a 1.5 layer like solution which has the limit of  $90^\circ$  and when  $\varphi \rightarrow \infty$  the maximum value of the profile decreases, providing a smaller change with altitude, and a more constant like viscosity with a classical angle of  $45^\circ$ .

The result from the linearly decreasing SUBC model is shown in Figure 10, where the values smoothly transition from  $90^\circ \rightarrow 45^\circ$  as  $\varphi$  increases. This plot resembles the exponential decaying model seen in Figure 5 and the 1.5 model as seen in Figure 2. Again, all angles are larger than the reference angle of  $45^\circ$ . The close resemblance to the exponential and 1.5 layer solutions makes sense since this solution is also a strictly decreasing solution.

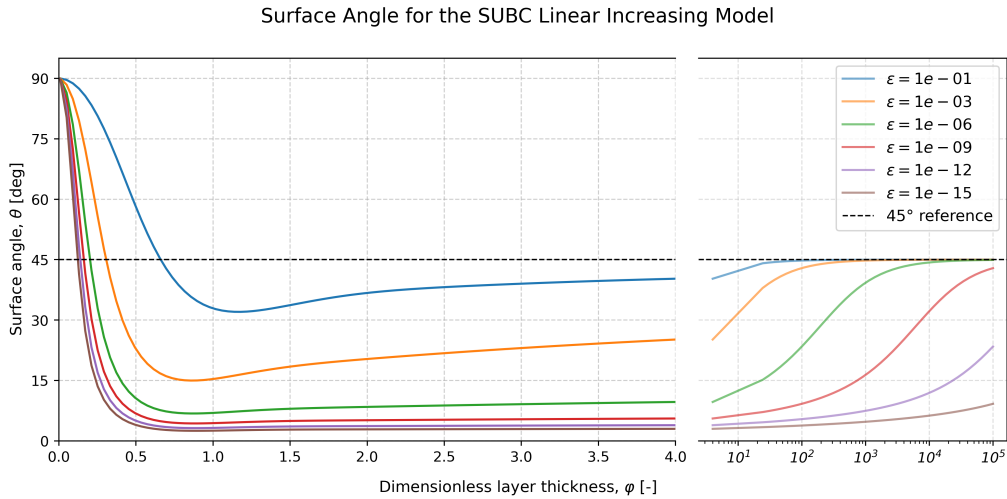


**Figure 10:** Numerically computed surface angles for decreasing viscosity profiles in the SUBC model. The angles are dependent on the dimensionless parameter  $\varphi = H/h_{EK}$ .

For the linearly increasing viscosity in Figure 11 a different result can be seen. The upper limit is again located at  $45^\circ$ , but at very large non-dimensional heights. For smaller background viscosities the limit occurs at many magnitudes higher then for the previous models due to a much larger  $1/\tilde{\nu}_\uparrow$ -factors.

The curve obtains values smaller then  $45^\circ$  outside of the limits discussed above. Since the limits are the extreme cases, this interior is the more interesting part of the result. The surface wind angle is smaller then  $45^\circ$  since a larger viscosity at a higher altitude means that the rotation of the wind is smaller near the surface and greater aloft. This forces the surface angles to become less then  $45^\circ$ . This can be compared to the two-layer model with a small lower viscosity  $\nu_1$  and small interface height  $H$ . Observing Figure 2, one can see that this scenario also yielded angles smaller than  $45^\circ$ , like in this figure.

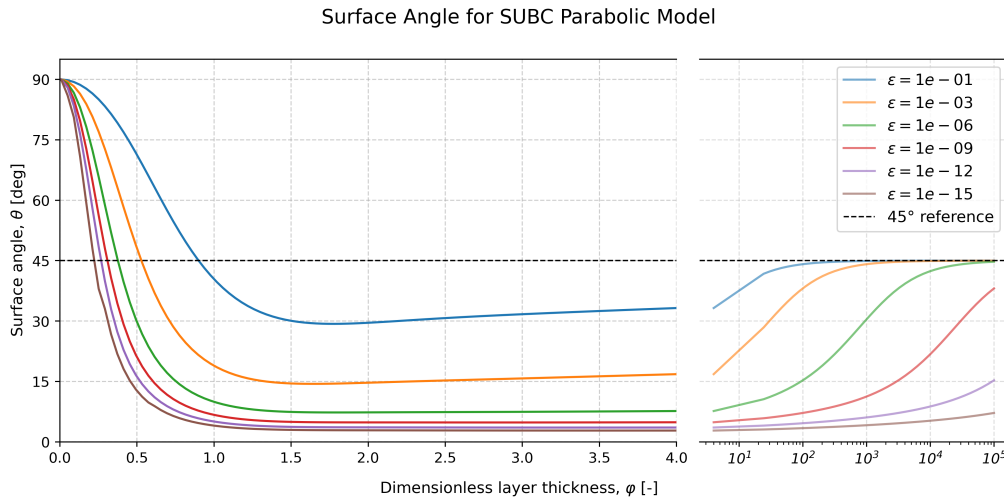
Another interesting observation is that, away from the limits, the surface wind angle is more sensitive to the background viscosity than to the non-dimensional height. If the background viscosity is much smaller then the maximal viscosity then a smaller  $\epsilon$  gives a huge  $1/\tilde{\nu}_\uparrow$  jump from the surface. In the theoretical limit when  $\epsilon \rightarrow 0$ , the initial jump would be infinite, and the surface angle would be zero.



**Figure 11:** Numerically computed surface angles for linearly increasing. The angles are dependent on the dimensionless parameter  $\varphi = H/h_{EK}$  and the background viscosity  $\epsilon$ .

The SUBC model with a parabolic viscosity profile gave Figure 12. Very similar behaviour to the increasing viscosity profile in Figure 11 can be observed. Since the viscosity jump from the surface is what determines the surface angle, the parabolic decrease aloft has no dramatic effect on the surface

angle. The two models differ slightly in that the parabolic curves cross the  $45^\circ$  later than the linearly increasing curves.



**Figure 12:** Numerically computed surface angles for a parabolic viscosity profile in the SUBC model. The angles are dependent on the dimensionless parameter  $\varphi = H/h_{\text{Ek}}$  and the background viscosity  $\epsilon$ .

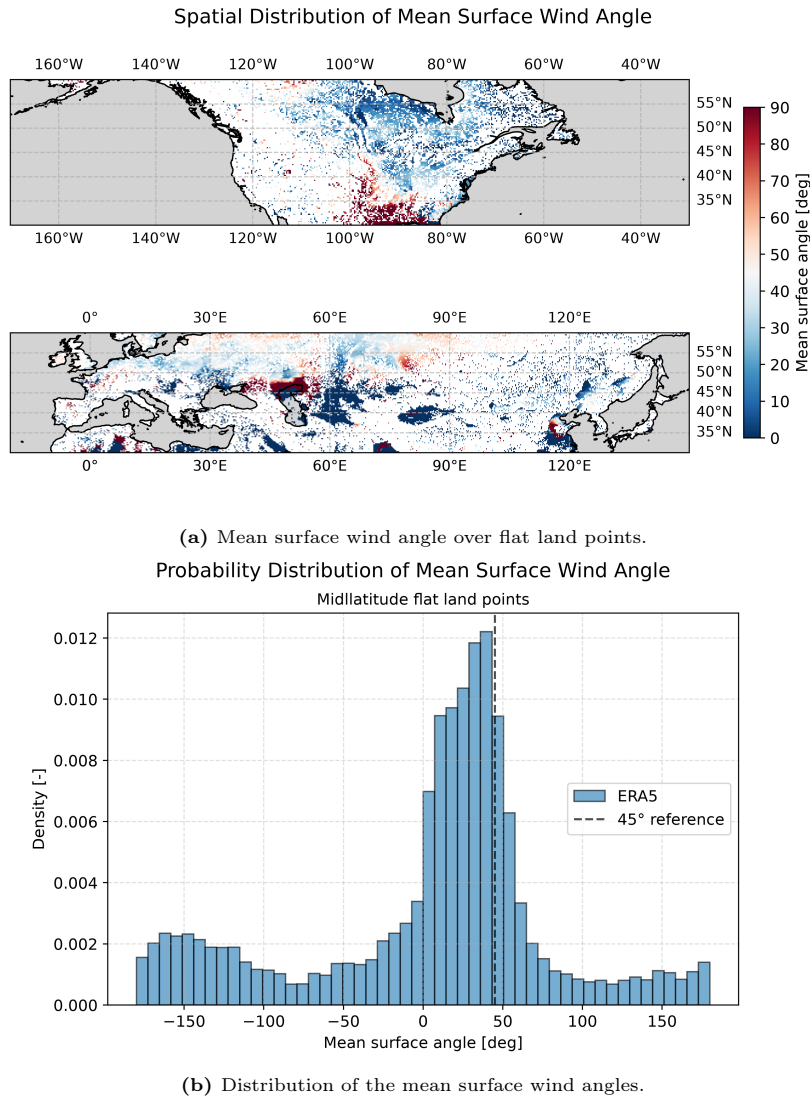
#### 4.1.3 The Ekman transport

Computing the transport with the surface vertical momentum flux as Equation 13 yielded exactly the same shapes as seen in Figure 2–12, but centred around  $135^\circ$  rather than  $45^\circ$ . The angle differences between the surface angles and the transport angle was thus always  $90^\circ$ , just like in the classical Ekman theory.

## 4.2 The ERA5 surface angles

24 535 data points were fit for analysis from the ERA5 dataset. These surface wind angles, due to the presence of a jet stream for flat land points in the northern midlatitudes, are shown in Figure 13. The geographical spread of the data points can be observed in Figure 13a, while Figure 13b shows the probability distribution. Figure 13b shows a wide range of surface angles, but with a clear peak between  $0^\circ$ – $60^\circ$ . Comparing this peak with the reference value of  $45^\circ$  shows that a clear majority of the surface angles are smaller than the reference, although a minority can be observed to be greater than the reference angle. The larger spread between  $-180^\circ$  to  $180^\circ$  may arise from other meteorological processes, such as the presence of a semi-permanent high- and low-pressure system, causing surface winds. Due to the smoothness of these values, these can be dismissed as outliers and the focus can be on the peak between  $0^\circ$ – $60^\circ$ .





**Figure 13:** Statistical analysis of the surface wind angles from ERA5 for flat land points located under a jet stream.

## 5 Discussion

From both analytical and numerical models, angles between  $0^\circ$  to  $90^\circ$  could be observed. Generally, surface wind angles above the classical value of  $45^\circ$  could be observed with strictly decreasing viscosities with altitude. The only case where this did not occur was for a small  $\varphi$ -region in the two-layer model, with a lowest value of  $43.4^\circ$ . The reasoning behind decreasing viscosities forcing rotation to occur at a lower height. As the viscosity is initially large but decreases with height, the spiral is forced to turn more at the lower altitudes. This has to occur in order to satisfy the geostrophic limit above. Thus, the surface angle becomes larger than the classical value for the decreasing viscosities with altitude.

For the increasing viscosities with altitude, the opposite could be observed with smaller surface wind angles than the classical value. For the two-layer model, only a small region had angles greater than the classical value. However, the largest value was only measured at  $46.6^\circ$ , which is close to the classical limit. The same logic follows to explain why this surface angle became smaller, as the explanation for why the surface angles became larger for the decreasing viscosities with height. If the viscosity increases with height, then the spiral has to turn slower near the surface in order to turn enough to reach the geostrophic limit aloft. And since the viscosity increases with altitude, more rotation can occur there, as

it is the shear due to the viscosity which forces this rotation.

The same reasoning can be used to explain the limits for the surface angles when the dimensionless layer thickness goes to infinity or zero. Since the dimensionless layer thickness is the ratio between the characteristic height and the Ekman height scale, it follows that when the ratio goes towards infinity, a constant viscosity is obtained, and the angles go to the classical value of  $45^\circ$ . Furthermore, when the ratio between the characteristic height and the Ekman height scale goes towards zero, one would obtain the extremes of the logic presented in the paragraphs above. Since the Coriolis force is perpendicular to the flow, a  $90^\circ$  surface angle will be obtained if the viscosity decreases with altitude, since the entire rotation has to occur at the surface. On the other hand, if the viscosity increases with height, then no part of the rotation can occur at the surface, and a parallel angle of  $0^\circ$  will be obtained at the surface. This could be explained more mathematically as when  $\varphi \ll 1$  the solution behaved approximately as a linear function, causing a shift in the phase of the surface stress. Because  $U(z) \approx \tau_0 z / \nu(0)$  close to the surface, this phase directly determines the surface wind angle, approaching  $90^\circ$  for decreasing viscosities and  $0^\circ$  for increasing viscosities.

Comparing the results from the theoretical models with the ERA5 data, a clear connection can be observed. The ERA5 data showed a clear majority of angles between  $0^\circ$  and  $45^\circ$ . As the atmosphere consists of a turbulent boundary layer viscosity generally increases with height, which connects the theoretical models to the observed data. However, a small portion of the data that had angles above  $45^\circ$ , although a boundary layer must be present. This can be attributed to the boundary layer having a small region where the viscosity decreases with height, or a strictly decreasing one producing slightly higher angles like in the two-layer model, which would allow surface angles above  $45^\circ$ . Although one should not forget that the ERA5 data are reanalysed data, and thus not pure observations, which also could explain this portion of the data. However, the bulk of the data is still in agreement with the theoretical models, supporting the findings in this article.

Comparing these results with previous works shows a general consistency. The general rule that strictly increasing viscosities with height produce surface angles below  $45^\circ$  and strictly decreasing viscosities with height produce surface angles above  $45^\circ$  can be compared to the previously mentioned earlier work. Ionescu-Kruse [2021] showed that by analysing two smoothly increasing polynomial viscosity profiles, only one of the two produced angles below  $45^\circ$ , with a quadratic profile producing angles above  $45^\circ$ . This was, however, corrected by Guan et al. [2023], who showed that the quadratic profile produced angles below  $45^\circ$ , agreeing with the findings in this article. The piecewise constant two-layer models by Dritschel et al. [2020] agreed with the findings here, that increasing viscosity yielded angles below  $45^\circ$  and vice versa. The perturbative studies by Bressan and Constantin [2019] and Constantin [2021] also support the findings in this article. Constantin [2021] highlighted that even strictly decreasing viscosities could yield angles below  $45^\circ$  and vice versa, like the two-layer model in this study. However, it seems like this can only occur for non-smoothly varying viscosities, as all the other models in this article and the previous works show that smoothly increasing viscosities yield angles below  $45^\circ$  and smoothly decreasing viscosities yield angles above  $45^\circ$ .

## 6 Conclusion

The two-layer model showed that strictly increasing viscosities with height mostly produced surface angles below  $45^\circ$ , while strictly decreasing viscosities mostly produced surface angles above  $45^\circ$ . The exponentially decaying model showed surface angles above  $45^\circ$ . The numerical model used a simplified upper boundary condition, which allowed for any viscosity profile to be studied. The linearly decreasing model gave surface angles above  $45^\circ$ , while the linearly increasing model and the parabolic model produced surface angles below, neglecting extremes caused by the new boundary condition. This showcased that the viscosity jump from the surface is what generally determines the surface angle, as increasing viscosities with height force the turning of the wind to occur aloft, while decreasing viscosities with height force the turning of the wind to occur near the surface. This was even true for the parabolic model, which had a decreasing viscosity aloft, giving angles below  $45^\circ$ . The analytical two-layer model could produce surface angles outside this rule, but only in a small region with lower magnitude angles.

From ERA5 data a clear majority of angles between  $0^\circ$  and  $45^\circ$  could be observed, supporting the atmosphere having an increasing viscosity near the surface, which agrees with the theory of the mixing length hypothesis and the boundary layer. This article has demonstrated how a combination of analytical, numerical, and reanalysed data can be used together to obtain a deeper understanding of the Ekman spiral and the effect of varying eddy viscosities with height. The results from this article can thus be used to further understand the Ekman spiral and thus get a deeper knowledge of the boundary layer in the atmosphere and ocean.

## Acknowledgment

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## A The Source Code

The source code for this article can be found on GitHub under:

<https://github.com/FredrikBergeliv/Ekman-Spiral-Project>

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